

Episodic Incident Response Agent

Experience-Augmented SRE Copilot via Four-Tier Engineering Memory

Domain: Engineering & DevOps

Active Spec: v2.4-PRODUCTION

Target SLA: MTTR < 2 Minutes

Classification: Engineering Blueprint

"The next incident should be easier because the agent remembers the last one."

When production is degraded or down, every minute translates to operational and financial loss. Retrieval-Augmented Generation (RAG) retrieves static docs, but retrieval is not experience. Our agent bridges this gap by persisting episodic execution traces, failure modes, and runbook efficacy across outage lifecycles.

1. PROBLEM STATEMENT & SCOPE BOUNDARIES

TARGET PERSONA & CORE BOTTLENECK	STRICT SCOPE BOUNDARY (DEFINITION OF DONE)
Primary Persona: On-call Site Reliability Engineers (SREs) and Systems Engineers under high cognitive pressure during critical outages. The RAG Bottleneck: Static RAG models recall documentation text but lack context on real outcomes. They cannot recall what commands failed 20 minutes ago, how pool limits evolved, or which mitigation proved counterproductive in historical post-mortems.	The prototype bounds execution strictly to one deterministic lifecycle loop: Incident Diagnosis: Triage from live telemetry & stack traces. Action Resolution: Human-gated remediation execution. Memory Writeback: Structured synthesis to the memory bank. <i>*Out of Scope:</i> Generic conversational chat, arbitrary ad-hoc scripts, and un-gated autonomous actions.

2. FOUR-TIER MEMORY TAXONOMY

Memory Class	Storage Semantics	Operational Payload / Concrete Example	SRE Value Proposition
EPISODIC	Chronological outage traces, failure timelines, runbook attempts.	INC-2026-0098: Experienced identical connection timeouts 14m post-deployment of checkout worker pods.	Prevents redundant root-cause discovery across rotating on-call shifts.
PROCEDURAL	Executable diagnostic trees, command recipes, validation order.	RB-CHK-07: Validate pg_stat_activity saturation before executing any rolling pod termination.	Enforces proven sequences and eliminates risky, trial-and-error CLI commands.
SEMANTIC	Generalized operational heuristics extracted from post-mortems.	MEM-042: Restarting checkout pods temporarily clears 5xx metrics but leaves postgres connection leaks unmitigated.	Blocks destructive "reboot-only" anti-patterns and systemic traps.
OUTCOME	Empirical remediation track records and success probabilities.	Pod restart succeeded 1/4 times (25%); pool_size config override succeeded 4/4 times (100%).	Ranks suggested actions by statistically proven historical utility.

MEMORY CONVERGENCE PRINCIPLE
During triage, episodic traces supply candidate runbooks (Procedural), filtered by empirical success rates (Outcome), and constrained by historical systemic invariants (Semantic). The agent behaves as a seasoned colleague rather than a raw search engine.

3. SYNTHETIC PRODUCTION ENVIRONMENT & SEEDED INCIDENT

Evaluation rigor requires a deterministic production failure context rather than stochastic mock strings. All interactions resolve against the following seeded topology:

TOPOLOGY & SERVICE ENVELOPE	ACTIVE OUTAGE DESCRIPTOR
Target Microservice: checkout-api (v2.14.2)	Incident Identifier: INC-2026-0142 P1 - CRITICAL
Service Owner: Payments Platform Squad	SLO Degradation: HTTP 5xx at 31.4% p99 > 4.200ms
Dependencies: payment-db (PostgreSQL 15), redis-cache	Bound Runbook: RB-CHK-07 (Checkout Saturation)
Triggering Event: Deployment DEP-8841 committed T-17m	Linked Memory Anchor: PM-2026-0098 / MEM-042

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[2026-09-28T01:14:02.108Z] CRITICAL checkout-api-pod-7db4f-8x2w1 [worker-3] psycopg2.OperationalError: pool timeout: server connection limit reached (50/50 connections active) at checkout/database/session.py:84 in acquire_connection() at checkout/services/payment.py:112 in process_transaction() [2026-09-28T01:14:05.412Z] healthz probe failed: HTTP 503 Service Unavailable (consecutive failures: 4)
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4. MULTI-STAGE LEARNING CURVE (THE DEMONSTRATION ARC)

The system evaluates agent performance across interactions to verify that memory accumulation directly drives diagnostic accuracy, suppresses failed mitigations, and shrinks recovery timelines.

Phase / State	Available Knowledge State	Observable Diagnostic Behavior	Mean Time to Resolve
Interaction 1 Baseline / Cold Start	Zero episodic history. Standard static markdown runbooks only.	Broad, cautious troubleshooting questionnaire. Suggests generic pod restart; requests manual DB inspection. Cannot determine root cause.	38 Minutes Baseline Benchmark
Interaction 5 Single History Match	Incident INC-2026-0098 + Post-Mortem PM-2026-0098.	Identifies pool exhaustion signature instantly. Flags pod restart as a transient failure anti-pattern; prescribes targeted max_connections patch.	9 Minutes -76% Reduction
Interaction 20 Multi-Incident Maturity	Cross-incident corpus (19 records, 6 post-mortems, 4 runbooks).	Differentiates DB saturation vs redis failure. Discards historically invalidated steps; generates verified patch script under 90s.	< 2 Minutes -94% Reduction

Empirical Demonstration Arc: Mean Time to Resolution (MTTR) Trajectory



5. STRICT 9-SECTION AGENT RESPONSE CONTRACT

To prevent ungrounded, hallucinated suggestions, every response emitted by the agent must conform to this 9-part structured schema:

1. Current Situation Active degradation profile, error impact (HTTP 5xx rate), service identity, and severity level (e.g., P1).	2. Current Evidence Literal quotes of live log lines, stack traces, saturation gauges, and failing health probe payloads.	3. Retrieved Memories Provenance links with similarity scores, Incident IDs (INC-xxx), and relevant post-mortem takeaways.
4. Strict Diagnosis Explicitly demarcated as either <i>[Hypothesis]</i> or <i>[Verified Root Cause]</i> with confidence rationale.	5. Ordered Action Plan Structured sequence: Verify → Mitigate → Diagnose → Durable Fix. No unstructured command blocks.	6. Rationale & Precedent Direct causal justification linking suggested commands to past post-mortem success/failure outcomes.
7. Uncertainty & Caveats Known unknowns, external factors (e.g., network partitions), and conditions that would invalidate the fix.	8. Human Approval Gate Mandatory explicit confirmation dialog before executing any mutating CLI or configuration command.	9. Memory Writeback Post-resolution synthesis formatting resolution data into an immutable record (MEM-xxx) for the bank.

6. PROTOTYPE DEFINITION OF DONE (SUBMISSION CHECKLIST)

All six evaluation criteria must pass for successful benchmark qualification:

✓ Interactive Incident Triggering: Evaluator UI allows initiating INC-2026-0142 or injecting simulated telemetry streams on demand.
✓ Live Evidence Grounding: Agent assessments cite concrete log output, active pod states, and database pool utilization metrics.
✓ Transparent Memory Provenance: Retrieved historical records display explicit similarity metrics, source incident IDs, and post-mortem links.
✓ Observable Multi-Stage Arc: Demonstrable progression from Cold Start (Interaction 1) to Targeted Mitigation (Interaction 5) and Rapid Resolution (Interaction 20).
✓ Enforced Human-in-the-Loop Gate: Consequential mutations (pod reboots, pool configuration writes) pause execution until manual approval is provided.
✓ Automated Memory Ingestion: Resolving the outage automatically compiles and registers a new MEM-xxx entity into the long-term memory bank.

Architectural Compliance Sign-Off		
SRE Governance Lead: [Approved]	Core Infrastructure Squad: [Approved]	Verification Status: READY FOR DEMO